

```
from airflow import DAG

from airflow.operators.python import PythonOperator

from datetime import datetime
```

```
# Task 1 -> Create Attendance File
```

```
def create_attendance_file():
```

```
    data = """Aarav,Present
Priya,Present
Rahul,Absent
Sneha,Present
Kiran,Absent
Ananya,Present
Vikram,Present
Meera,Absent
Farhan,Present
Divya,Present"""
```

```
    with open("/tmp/attendance.txt", "w") as file:
```

```
        file.write(data)
```

```
    print("Attendance file created successfully")
```

Task 2 -> Read Attendance File

```
def read_attendance_file():
```

```
    with open("/tmp/attendance.txt", "r") as file:
```

```
        records = file.readlines()
```

```
    print("Attendance Records:")
```

```
    for record in records:
```

```
        print(record.strip())
```

Task 3 -> Count Total Students

```
def count_total_students(ti):
```

```
    with open("/tmp/attendance.txt", "r") as file:
```

```
        records = file.readlines()
```

```
    total_students = len(records)
```

```
    print(f"Total Students = {total_students}")
```

```
    ti.xcom_push(key="total_students", value=total_students)
```

Task 4 -> Count Present Students

```
def count_present_students(ti):
```

```
    with open("/tmp/attendance.txt", "r") as file:
```

```
        records = file.readlines()
```

```
    present_students = 0
```

```
    for record in records:
```

```
        name, status = record.strip().split(",")
```

```
        if status == "Present":
```

```
            present_students += 1
```

```
    print(f"Present Students = {present_students}")
```

```
    ti.xcom_push(key="present_students", value=present_students)
```

Task 5 -> Count Absent Students

```
def count_absent_students(ti):
```

```
    with open("/tmp/attendance.txt", "r") as file:
```

```
        records = file.readlines()
```

```
    absent_students = 0
```

```
    for record in records:
```

```
        name, status = record.strip().split(",")
```

```
        if status == "Absent":
```

```
            absent_students += 1
```

```
    print(f"Absent Students = {absent_students}")
```

```
    ti.xcom_push(key="absent_students", value=absent_students)
```

Task 6 -> Calculate Attendance Percentage

```
def calculate_attendance_percentage(ti):
```

```
    total_students = ti.xcom_pull(  
        task_ids="count_total_students",  
        key="total_students"  
    )
```

```
    present_students = ti.xcom_pull(  
        task_ids="count_present_students",  
        key="present_students"  
    )
```

```
    percentage = (present_students / total_students) * 100
```

```
    print(f"Attendance Percentage = {percentage}%")
```

```
    ti.xcom_push(key="attendance_percentage", value=percentage)
```

Bonus Task -> List Absent Students

```
def list_absent_students():
```

```
    with open("/tmp/attendance.txt", "r") as file:
```

```
        records = file.readlines()
```

```
    print("Absent Students List")
```

```
    for record in records:
```

```
        name, status = record.strip().split(",")
```

```
        if status == "Absent":
```

```
            print(name)
```

Task 7 -> Generate Attendance Report

```
def generate_attendance_report(ti):
```

```
    total_students = ti.xcom_pull(  
        task_ids="count_total_students",  
        key="total_students"  
    )
```

```
    present_students = ti.xcom_pull(  
        task_ids="count_present_students",  
        key="present_students"  
    )
```

```
    absent_students = ti.xcom_pull(  
        task_ids="count_absent_students",  
        key="absent_students"  
    )
```

```
    percentage = ti.xcom_pull(  
        task_ids="calculate_attendance_percentage",  
        key="attendance_percentage"  
    )
```

```
if percentage >= 75:
    status = "Good"
else:
    status = "Needs Improvement"

report = f"""
Daily Attendance Report

Total Students = {total_students}

Present Students = {present_students}

Absent Students = {absent_students}

Attendance Percentage = {percentage}%

Status = {status}
"""

with open("/tmp/attendance_report.txt", "w") as file:
    file.write(report)

print("Attendance report generated successfully")
```

```
# DAG Definition
```

```
with DAG(
```

```
    dag_id="daily_attendance_processor_dag",
```

```
    start_date=datetime(2025, 1, 1),
```

```
    schedule=None,
```

```
    catchup=False
```

```
) as dag:
```

```
# Create Tasks
```

```
task1 = PythonOperator(
```

```
    task_id="create_attendance_file",
```

```
    python_callable=create_attendance_file
```

```
)
```

```
task2 = PythonOperator(
```

```
    task_id="read_attendance_file",
```

```
    python_callable=read_attendance_file
```

```
)
```

```
task3 = PythonOperator(  
    task_id="count_total_students",  
    python_callable=count_total_students  
)
```

```
task4 = PythonOperator(  
    task_id="count_present_students",  
    python_callable=count_present_students  
)
```

```
task5 = PythonOperator(  
    task_id="count_absent_students",  
    python_callable=count_absent_students  
)
```

```
task6 = PythonOperator(  
    task_id="calculate_attendance_percentage",  
    python_callable=calculate_attendance_percentage  
)
```

```
task7 = PythonOperator(  
    task_id="list_absent_students",  
    python_callable=list_absent_students  
)
```

```
task8 = PythonOperator(  
    task_id="generate_attendance_report",  
    python_callable=generate_attendance_report
```

)

Task Dependencies

task1 >> task2 >> task3 >> task4 >> task5 >> task6 >> task7 >> task8